

VLSI Design & Practice 05

22-bit Carry Save Multiplier (RCA, SRCSA, 2-stage Pipelined)

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Hyunseok Jung
hsj_@korea.ac.kr

Hyeon-Gwon Kim
gusrnjs48@korea.ac.kr

Jangseok Yu
jangseok@korea.ac.kr

Minjong Yoo
yoom1144@korea.ac.kr

Contents

- 1. 22×22 Carry Save Multiplier (RC adder)**
 - Design Carry Save Multiplier with Gate level description
- 2. 22×22 Carry Save Multiplier (SRCS adder)**
 - Modify the previous Carry Save Multiplier
- 3. 2-stage Pipelined 22×22 Carry Save Multiplier (RC adder)**
 - Design Carry Save Multiplier with Gate level description
- 4. Simulation & Synthesis Results**
 - Check the functionality and Add the flip-flops
 - Measure the critical path delay & circuit area
 - Check the experimental results (max. clock cycle, area, power)

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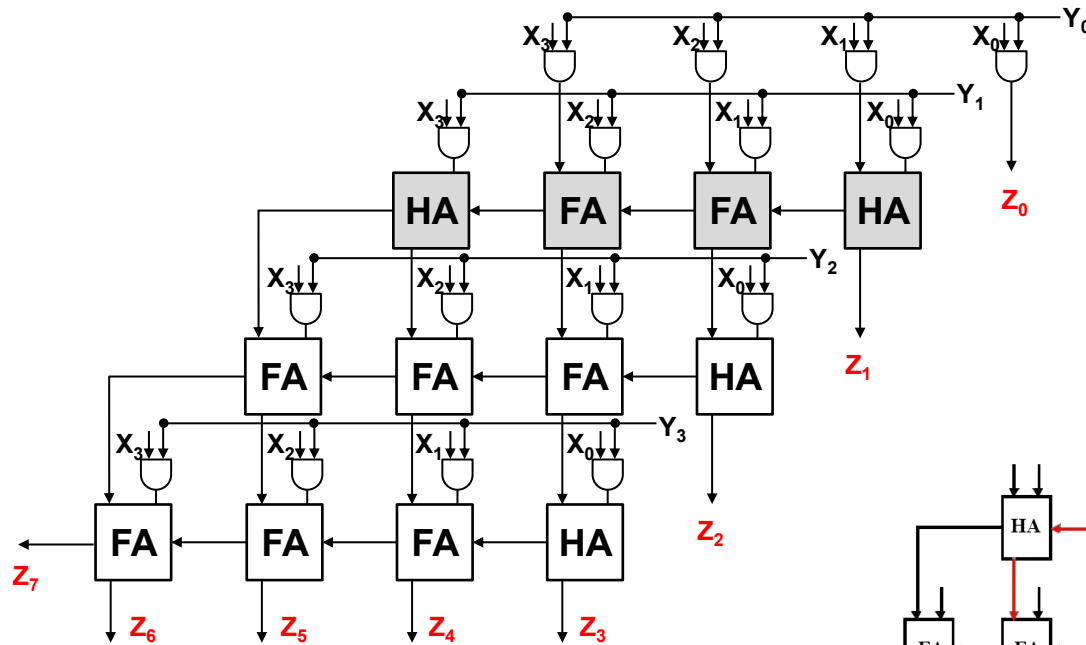
- Design Carry Save Multiplier with Gate level description

4. Simulation & Synthesis Results

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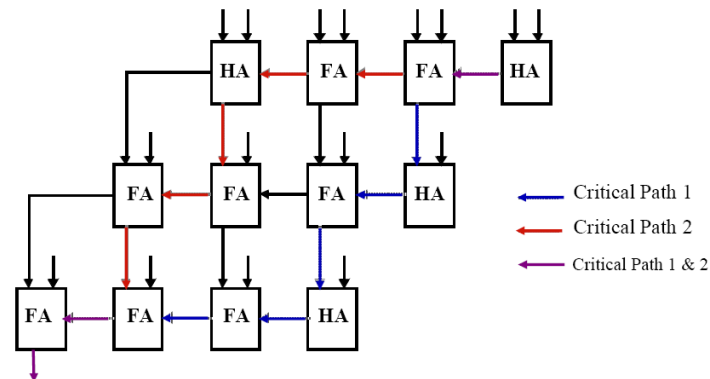
Array Multiplier

Example of 4×4 Array Multiplier



$$\begin{array}{r}
 1010 \\
 10\boxed{1}1 \\
 \hline
 1010 \\
 \boxed{1}010 \\
 0000 \\
 + 1010 \\
 \hline
 01101110
 \end{array}$$

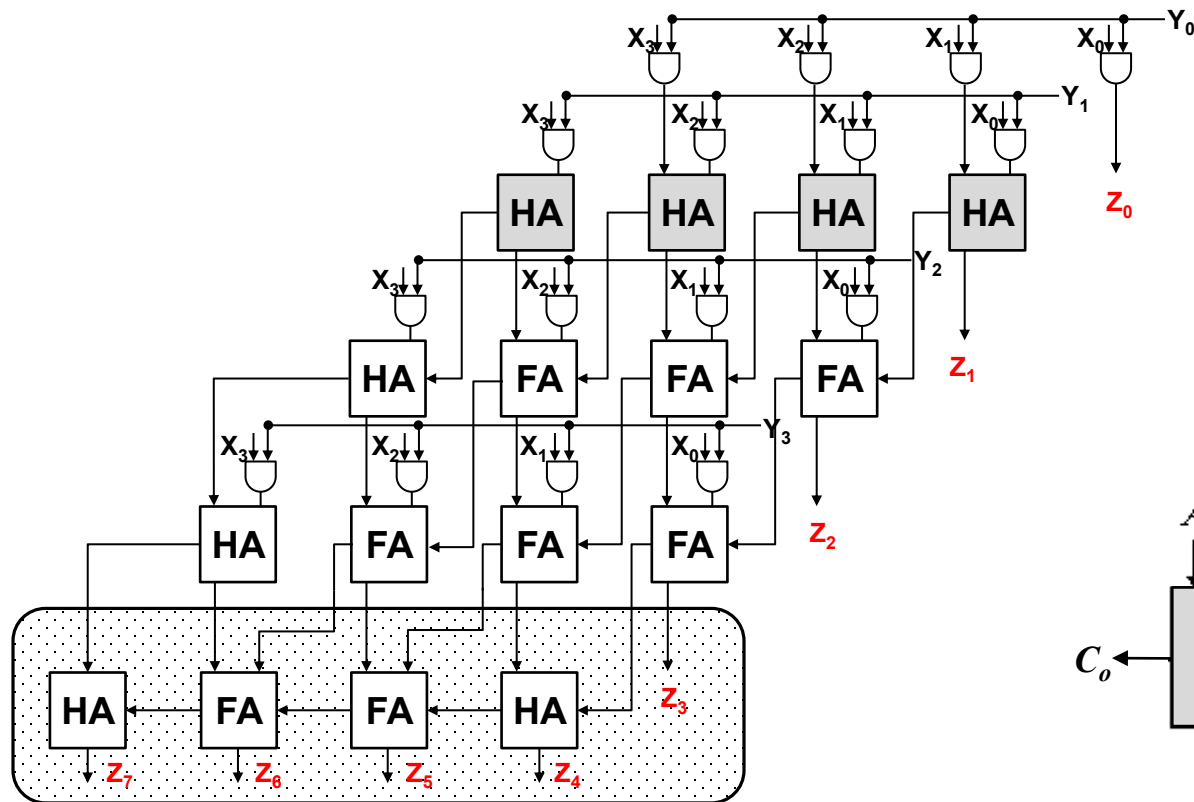
Let assume the delay
: HA $\cong$ FA



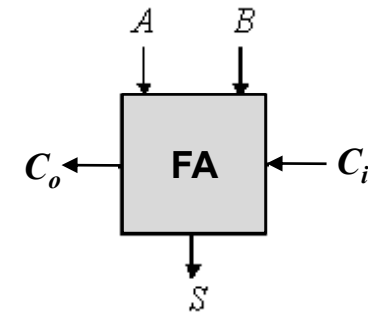
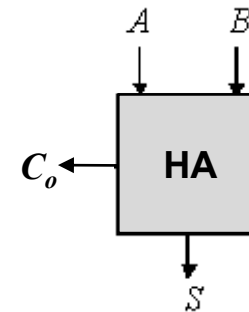
4×4 Carry Save Multiplier

Design 22×22 Carry Save Multiplier (RC Adder in the last stage)

Ex) 4×4 Carry Save Multiplier



$$\begin{array}{r}
 \begin{array}{cccc}
 & 1 & 0 & 1 & 0 \\
 & 1 & 0 & 1 & 1 \\
 \hline
 & 1 & 0 & 1 & 0 \\
 & 1 & 0 & 1 & 0 \\
 & 0 & 0 & 0 & 0 \\
 + & 1 & 0 & 1 & 0 \\
 \hline
 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0
 \end{array}
 \end{array}$$

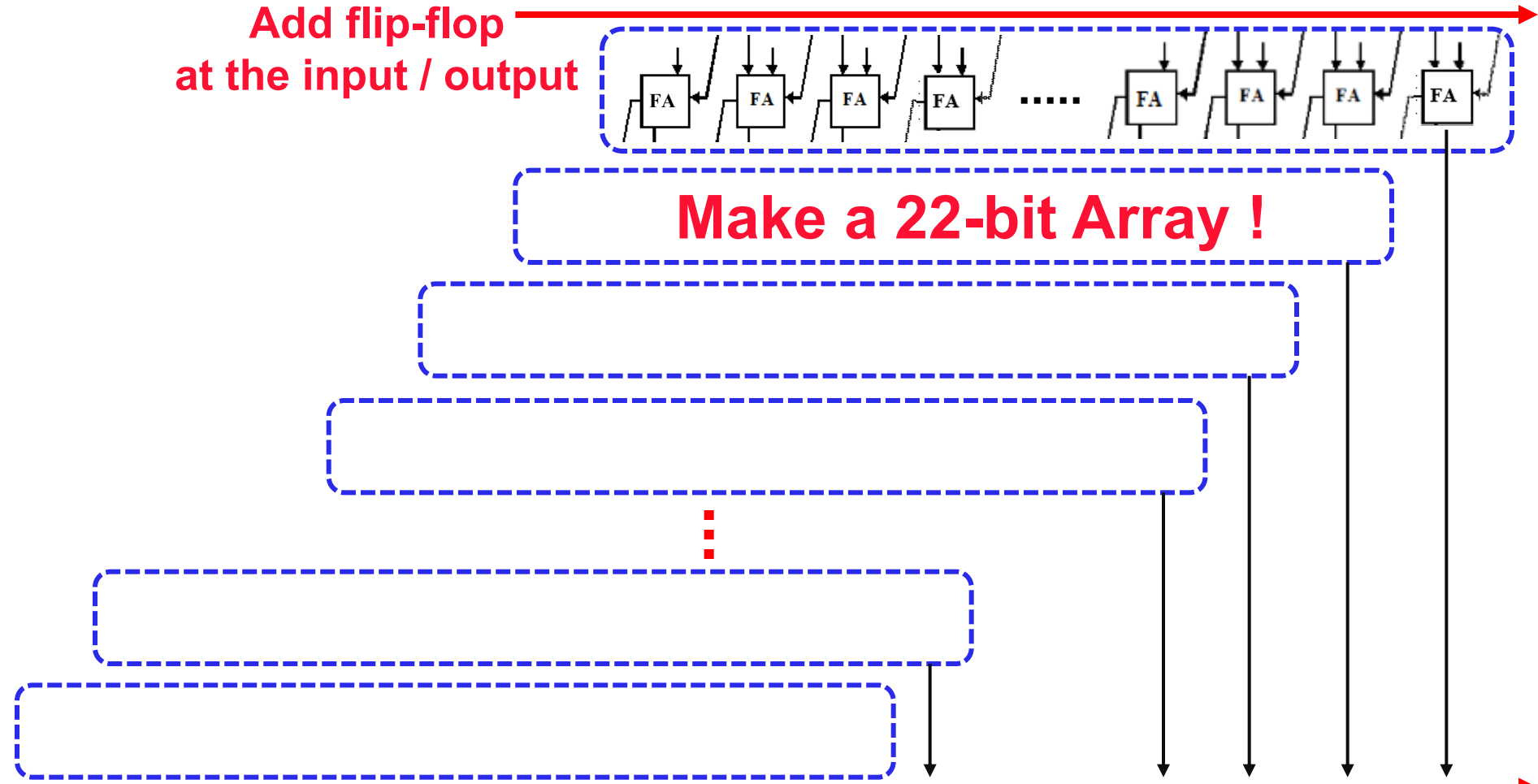


Ripple Carry Adder in the last Stage (Vector Merging Adder)

22×22 Carry Save Multiplier

Design 22×22 Carry Save Multiplier (Ripple Carry Adder in the last stage)

Add flip-flop
at the input / output



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- Design Carry Save Multiplier with Gate level description

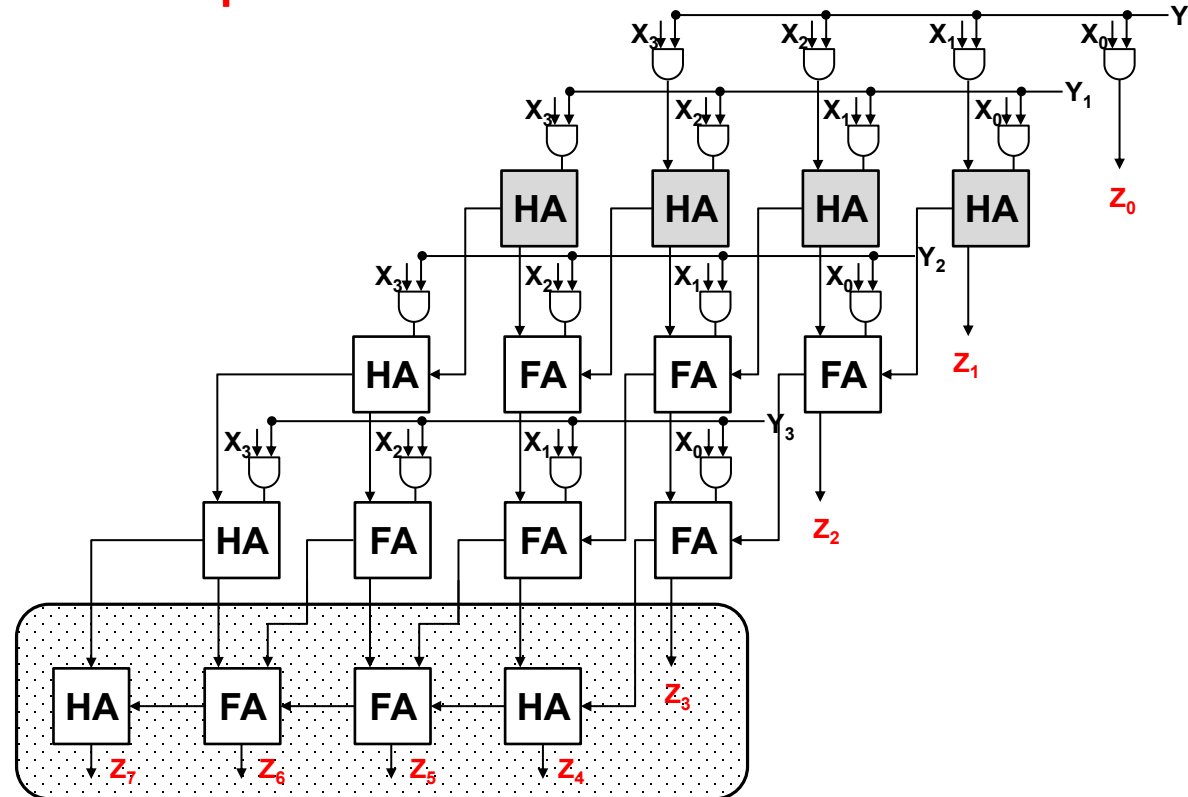
4. Simulation & Synthesis Results

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- Measure the critical path delay & circuit area
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22×22 Carry Save Multiplier

Design 22×22 Carry Save Multiplier (Square Root Carry Select adder)

Ex) 4×4 Carry Save Multiplier

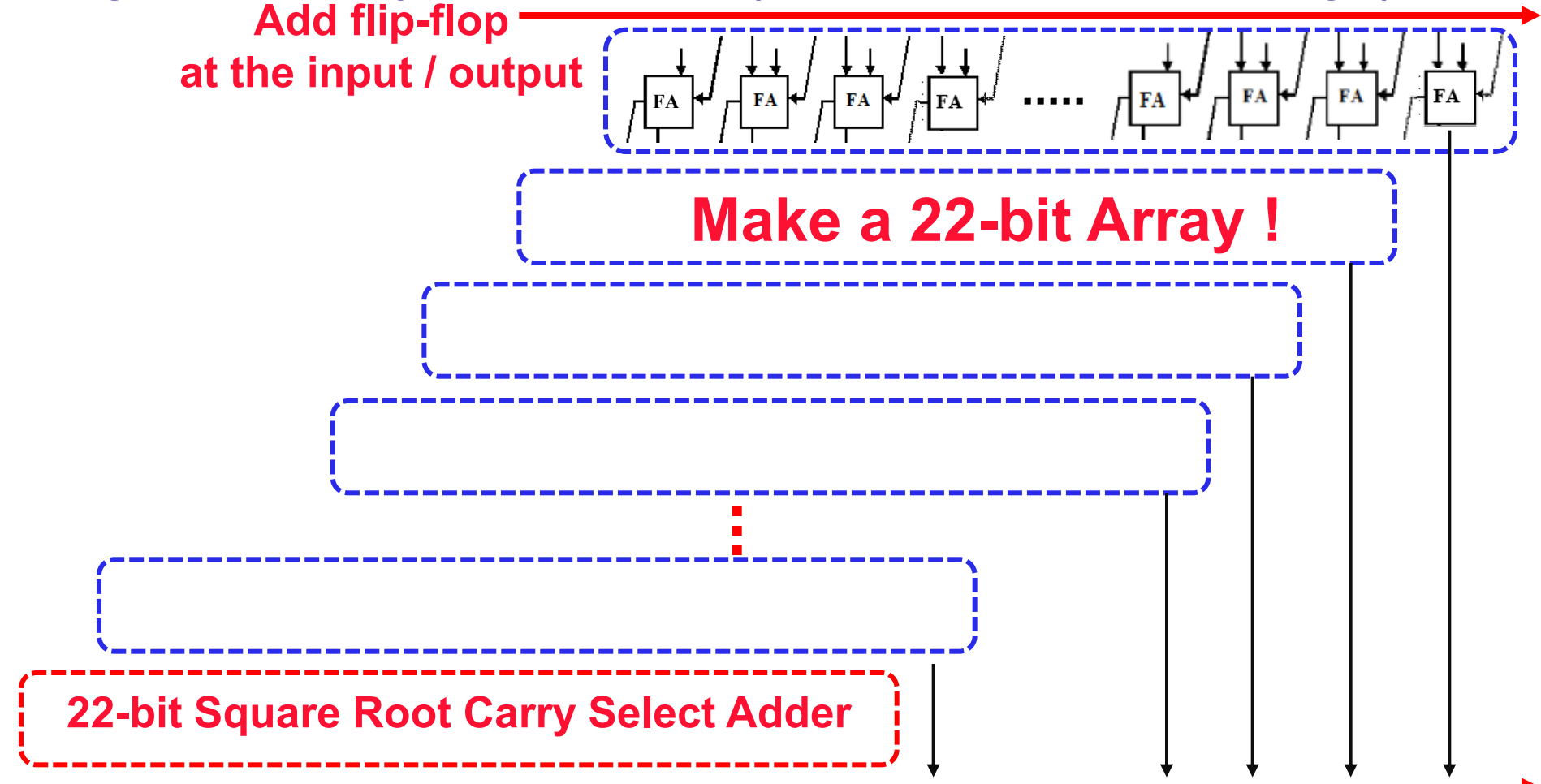


Last stage is still a carry propagation adder !!

22×22 Carry Save Multiplier

Design 22×22 Carry Save Multiplier (SRCS Adder in the last stage)

Add flip-flop
at the input / output



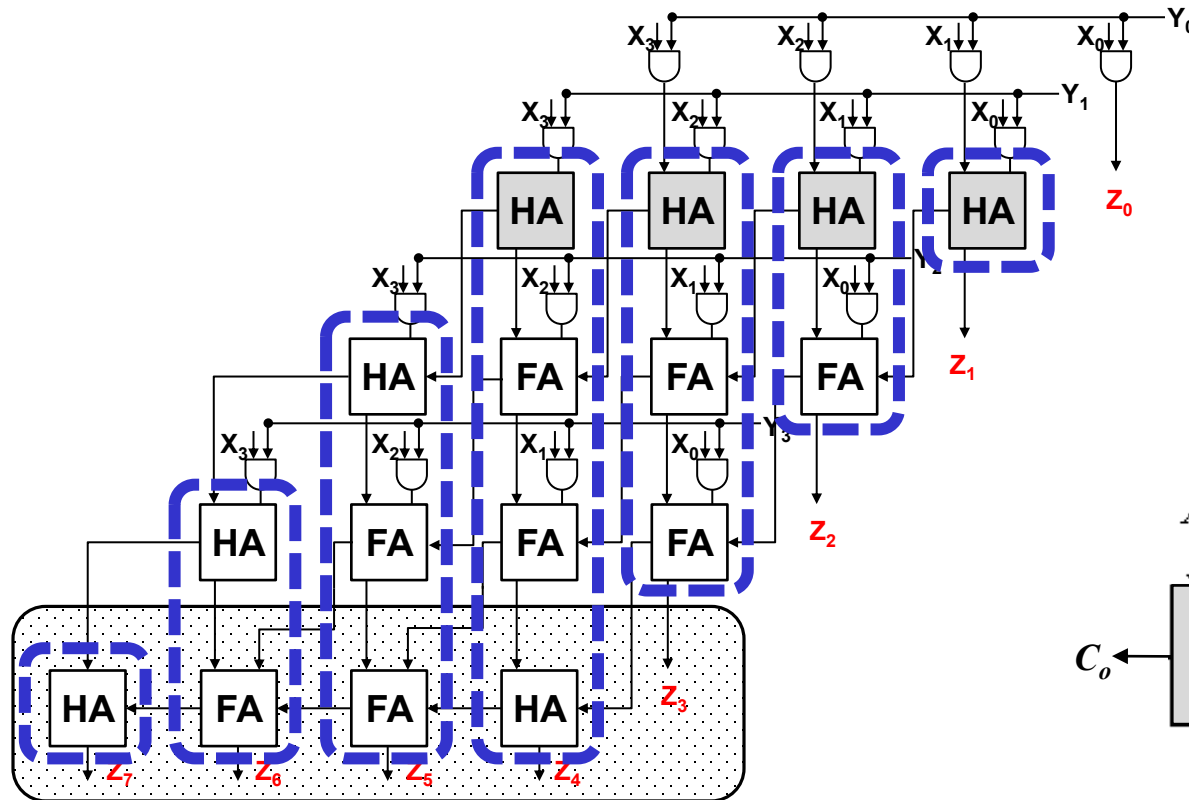
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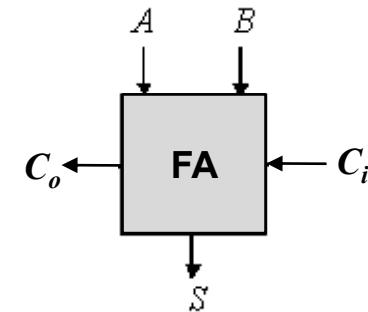
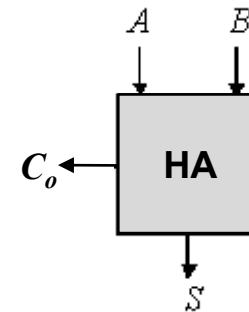
2-stage Pipelined Carry Save Multiplier

Design 22×22 Carry Save Multiplier (RC Adder in the last stage)

Ex) 4×4 Carry Save Multiplier



$$\begin{array}{r}
 \begin{array}{cccc}
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 1 \\
 \hline
 1 & 0 & 1 & 0 \\
 1 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 \\
 + & 1 & 0 & 1 & 0 \\
 \hline
 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0
 \end{array}
 \end{array}$$

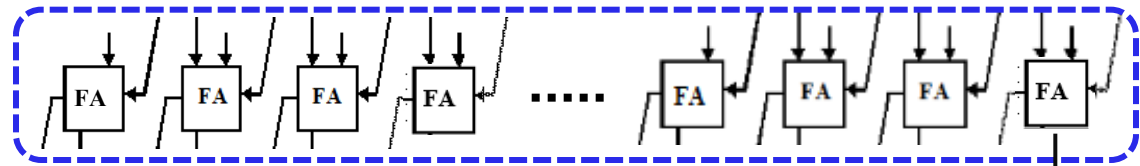


Ripple Carry Adder in the last Stage (Vector Merging Adder)

2-stage Pipelined Carry Save Multiplier

Design 22×22 Carry Save Multiplier (Ripple Carry Adder in the last stage)

Add flip-flop
at the input / output



Make a 22-bit Array !

Add 2-stage
pipelining flip-flop

Why?
Check paths

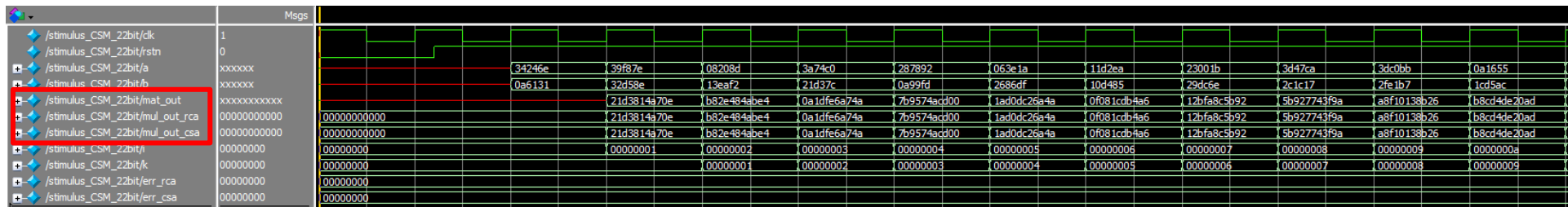
22-bit Ripple Carry Adder !

Contents

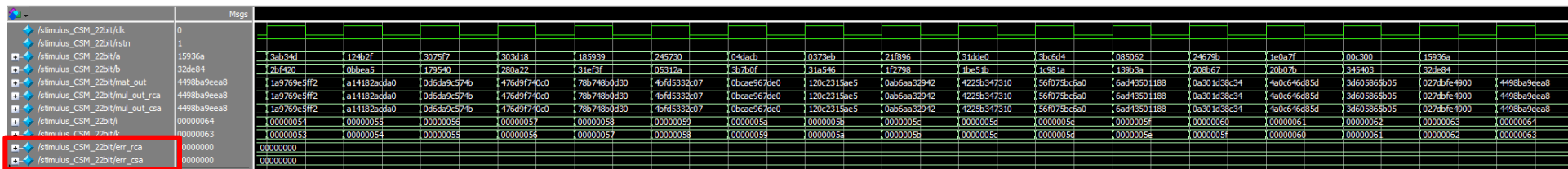
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Functional Verifications

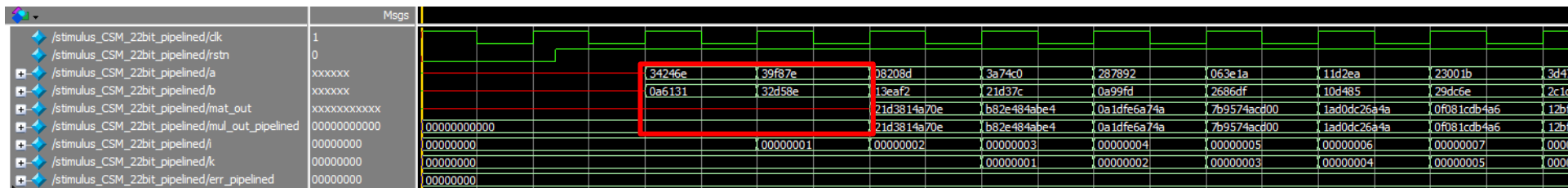
• Simulation results



- Check outputs of two multipliers and Matlab results are 'same'

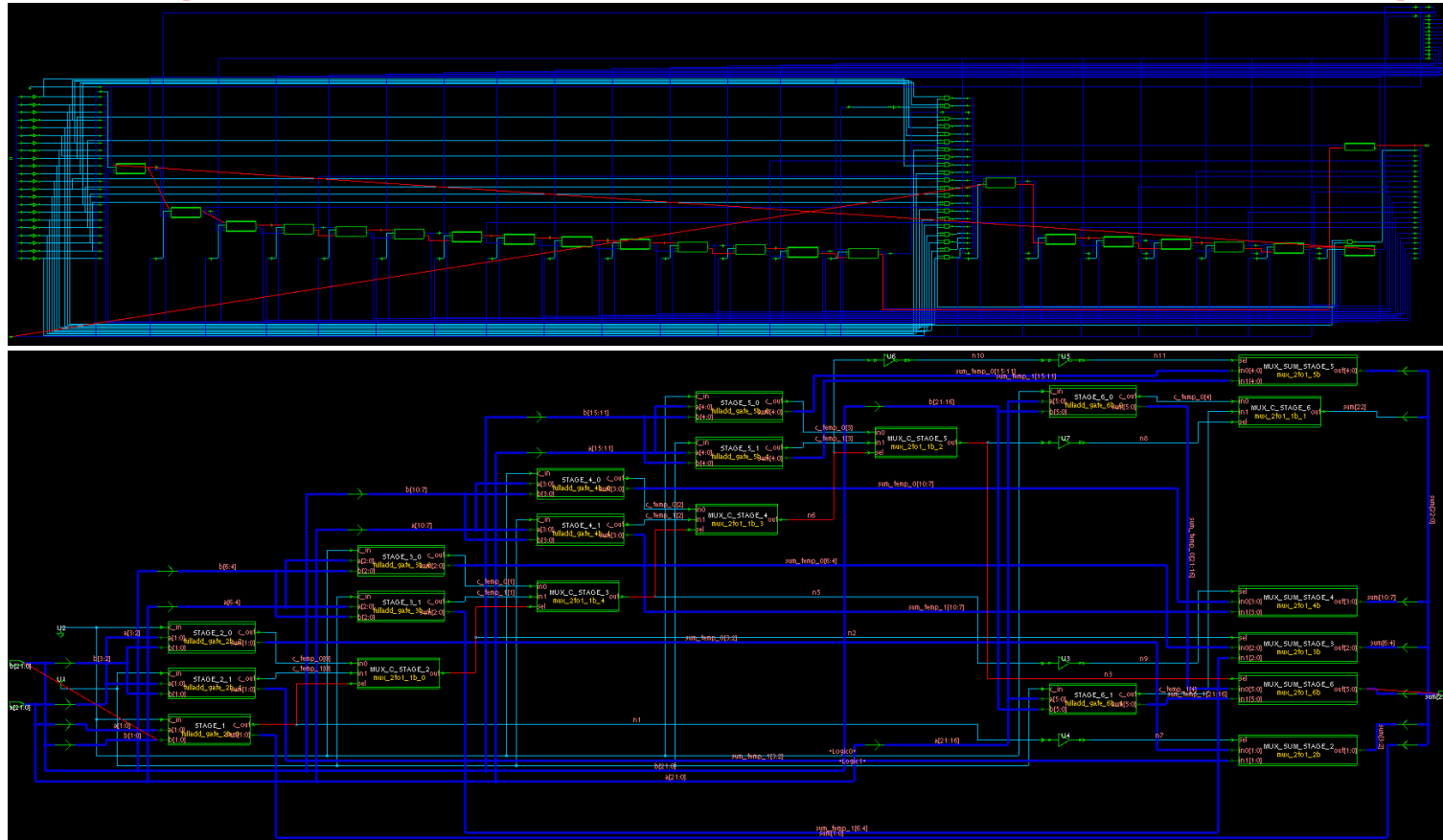


- Check the value of error counting number equals 'zero'



- Check the delay for 2-stage pipelined multiplier

- **Synthesis results** Carry Select Multipiler with SRCS adder
 - Check positions and names of each modules and find critical paths.



Check the synthesis results

- Critical path delay, Circuit area & Power

report_timing

RCadder		
clock clk (rise edge)	13.40	13.40
clock network delay (ideal)	0.00	13.40
D_OUT/q_reg[43]/CK (fd1qd4_hd)	0.00	13.40 r
library setup time	-0.08	13.32
data required time		13.32

data required time		13.32
data arrival time		-13.32

slack (MET)		0.00

SRCSadder		
clock clk (rise edge)	10.20	10.20
clock network delay (ideal)	0.00	10.20
D_OUT/q_reg[40]/CK (fd1qd1_hd)	0.00	10.20 r
library setup time	-0.11	10.09
data required time		10.09

data required time		10.09
data arrival time		-10.09

slack (MET)		0.00

report_area

RCadder	
Number of ports:	90
Number of nets:	178
Number of cells:	4
Number of references:	4
Combinational area: 9959.002762	
Noncombinational area: 497.333187	
Net Interconnect area: 56.001698	
Total cell area: 10456.335938	
Total area: 10512.337635	

SRCSadder	
Number of ports:	90
Number of nets:	178
Number of cells:	4
Number of references:	4
Combinational area: 10484.003137	
Noncombinational area: 504.666555	
Net Interconnect area: 54.447220	
Total cell area: 10988.669922	
Total area: 11043.117142	

report_power

RCadder	
Global Operating Voltage = 1.05	
Power-specific unit information :	
Voltage Units = 1V	
Capacitance Units = 1.000000pf	
Time Units = 1ns	
Dynamic Power Units = 1mW (derived from V,C,T units)	
Leakage Power Units = 1mW	
Cell Internal Power = 1.0478 mW (62%)	
Net Switching Power = 636.7448 uW (38%)	

Total Dynamic Power = 1.6846 mW (100%)	
Cell Leakage Power = 34.4576 uW	

SRCSadder	
Global Operating Voltage = 1.05	
Power-specific unit information :	
Voltage Units = 1V	
Capacitance Units = 1.000000pf	
Time Units = 1ns	
Dynamic Power Units = 1mW (derived from V,C,T units)	
Leakage Power Units = 1mW	
Cell Internal Power = 1.3695 mW (64%)	
Net Switching Power = 777.6013 uW (36%)	

Total Dynamic Power = 2.1471 mW (100%)	
Cell Leakage Power = 35.6985 uW	

Check results and Submit report

- **Check the synthesis result**
 - **Show waveforms & synthesis results (schematic)**
 - **Show timing, area, power report & .v file**
 - **Check the highlighted critical path**
- **Submit result report**
 - **Show the timing diagram and the above synthesis results**
 - **Results analysis**
- **Due : Before the next class to your group TA**
(담당 조교에게 제출 바랍니다)